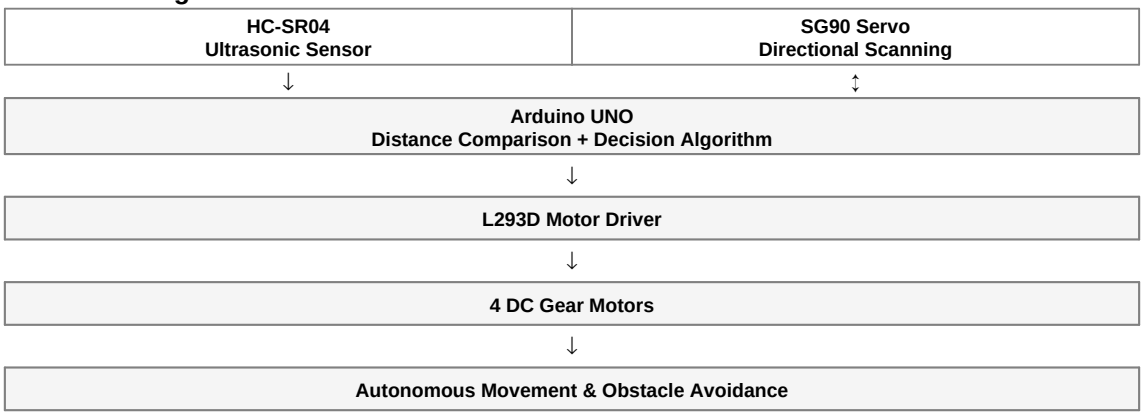
SAOANA – Smart Autonomous Obstacle Avoidance & Navigation Robot

AURA 2026 – Abstract Idea | One-Page Write-up

1. Methodology

SAOANA is a low-cost autonomous mobile robot designed to detect obstacles and choose a safer path without continuous human control. The robot uses an Arduino UNO as its main controller, an HC-SR04 ultrasonic sensor for distance measurement, an SG90 servo motor for directional scanning, an L293D motor driver, and four DC gear motors for movement. During normal operation, the ultrasonic sensor continuously checks the distance in front of the robot. When the path is clear, the robot moves forward. If an obstacle comes within a predefined safety distance, the robot stops and the servo rotates the sensor toward the left and right sides. The Arduino records both distances and compares the available space. It then commands the motor driver to move the robot toward the side having greater clearance. If both sides are blocked, the robot reverses for a short time and changes direction to search for another path. This process is repeated continuously, allowing the robot to navigate around obstacles using a simple distance-based decision algorithm. The system is designed to be easy to understand, assemble, test, and improve, making it suitable for a student-level robotics prototype.

2. Proposed Block Diagram



3. Proposed Outcome

The proposed outcome is a working four-wheel autonomous robot capable of detecting obstacles, scanning alternative directions, selecting a suitable path, and recovering from simple dead-end situations. The prototype will demonstrate how sensors, embedded programming, motor control, and mechanical movement can work together to achieve autonomous navigation using affordable components. During testing, the robot will be placed on a small obstacle track containing straight paths, side obstacles, and dead-end sections. Its movement and path selection will be observed under different conditions. The project can also serve as a basic platform for educational robotics and can be further developed for indoor service, delivery, inspection, and other autonomous robotic applications.

4. Innovation

The main innovation is the use of servo-based directional scanning and distance comparison instead of a fixed left- or right-turn rule. This gives the robot a more logical response to changing obstacles. A dead-end recovery mechanism is also included so that the robot can reverse and search for an alternative path when necessary. The design focuses on practical navigation with simple, affordable hardware rather than depending on expensive sensors or advanced computing.

Approximate write-up word count: 367 words

